

# Animal behavior analysis in python

Day 4 MoSeq introduction

# **Characterizing the structure of mouse behavior using Motion Sequencing**

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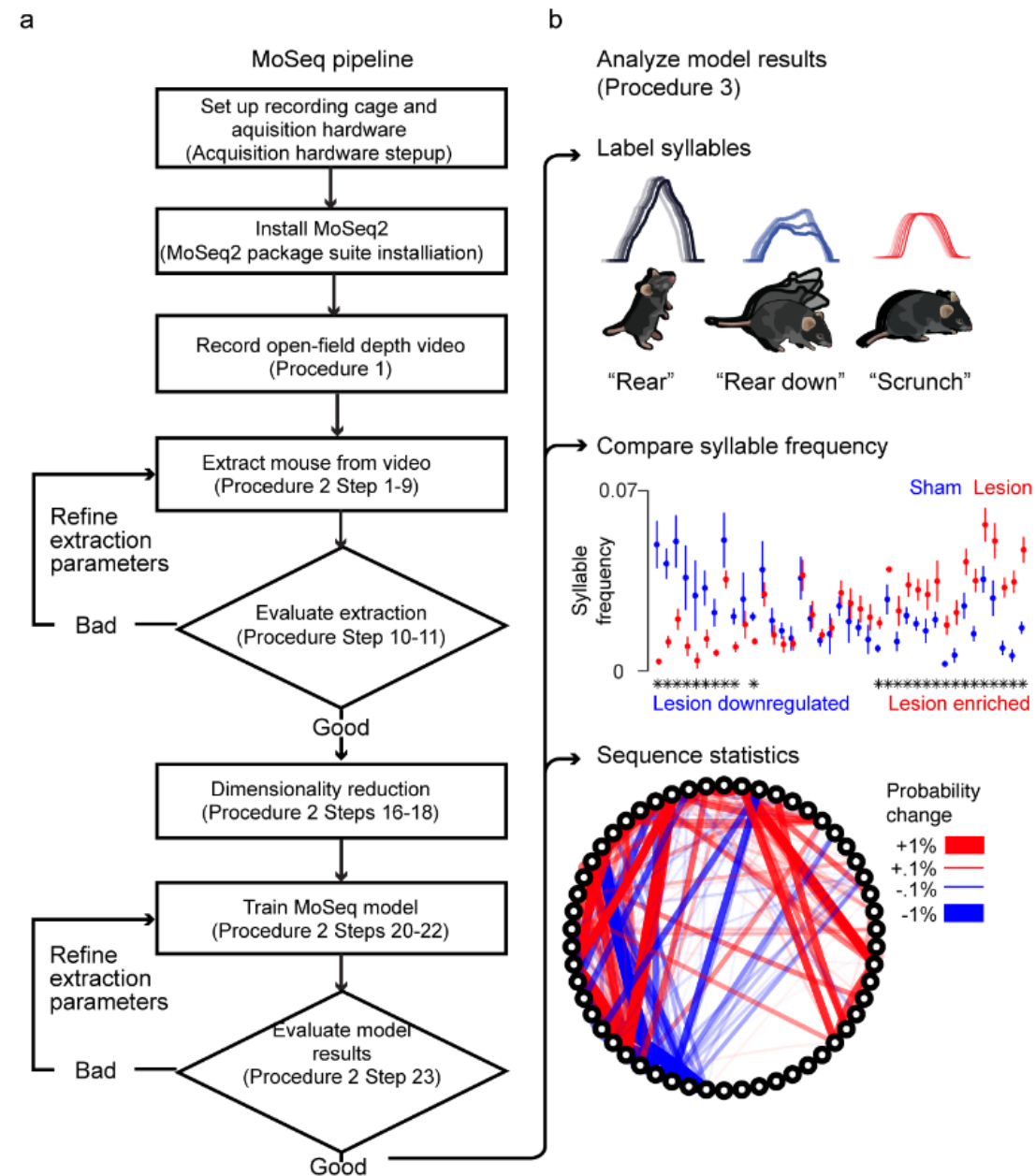
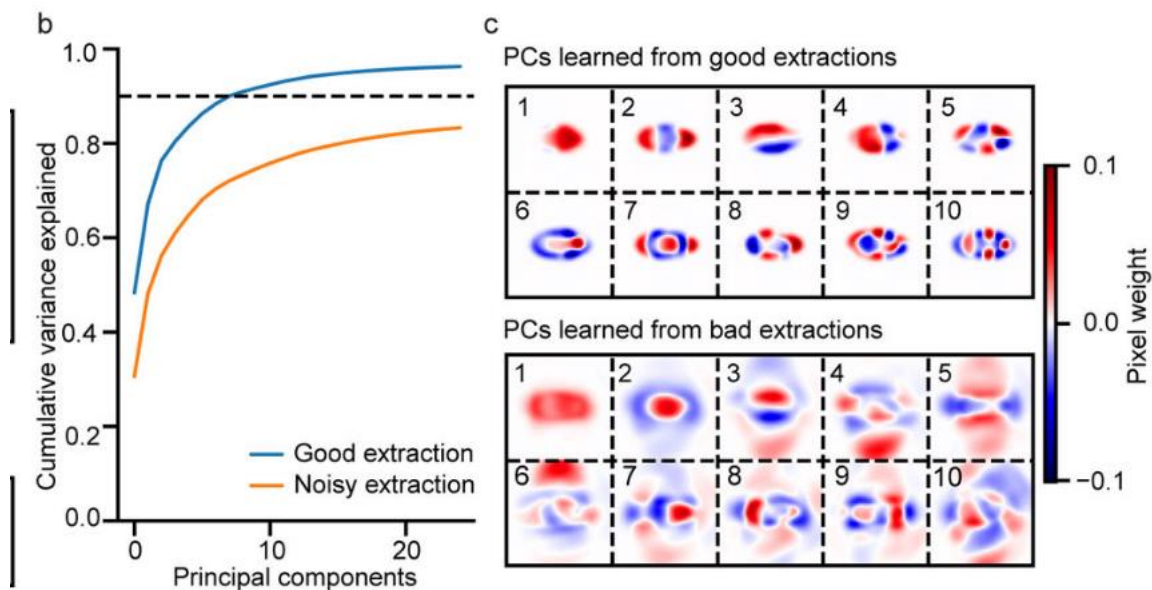
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# Moseq

From Spontaneous mouse behavior to  
(series of elemental modules)  
called “syllables”.

## 1. PCA



## 2. Change point detection

- describe moments in time when one action transitions into another, as determined by a model-free changepoints detection algorithm

selection of a parameter called “kappa” later in the protocol, which determines the global duration distribution of syllables. Typically, the distribution of changepoint durations in a mouse experiment is smooth, right-skewed, and peaks around 0.3 seconds (Fig. 7b).

### 3. a preliminary set of fit AR-HMM model(s)

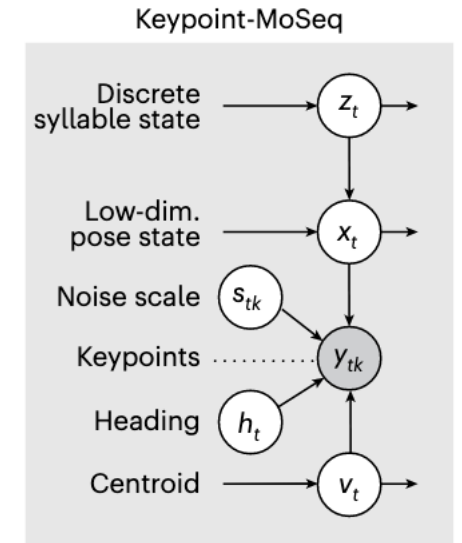
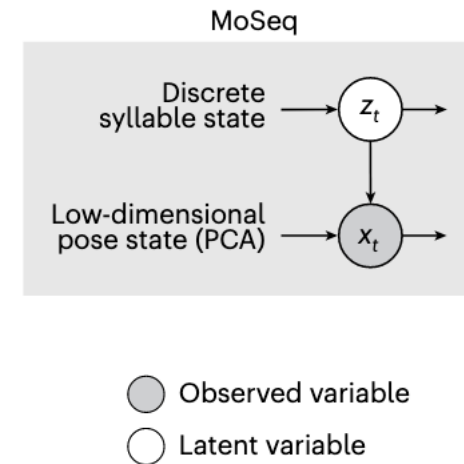
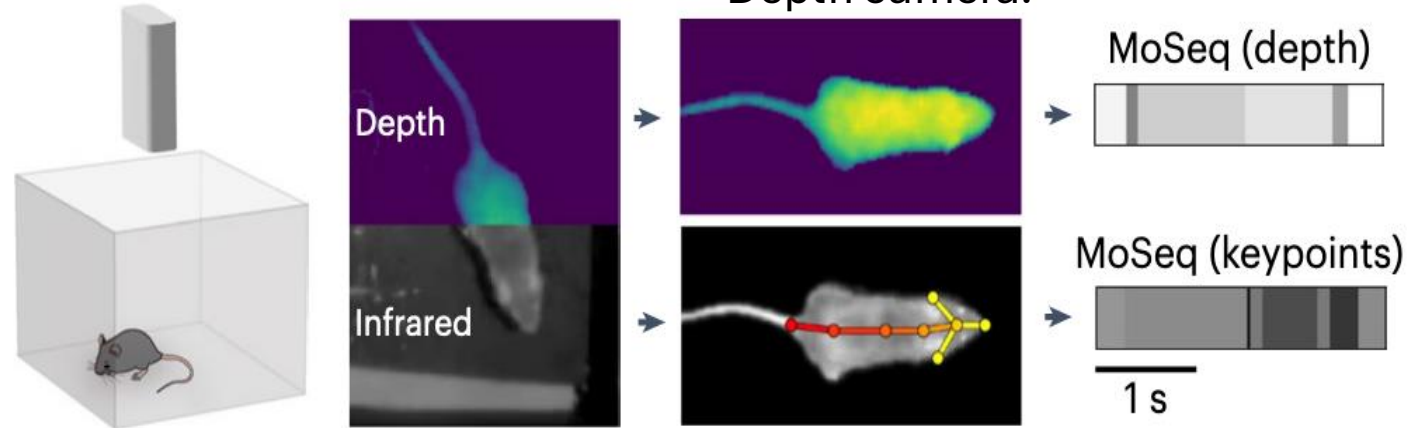
- Set Model Parameter cell and the Train Model cell

kappa: this value determines the duration of each syllable. The mean syllable durations increase as kappa value increases (Fig. 7c). By default, kappa is set to “none,” which means that kappa will be equal to the number of frames in the dataset. We have empirically observed that for most datasets a kappa value equal to the number of frames will yield a syllable duration distribution that is well matched to the mean of the changepoints distribution. Therefore, “none” should return appropriate results.

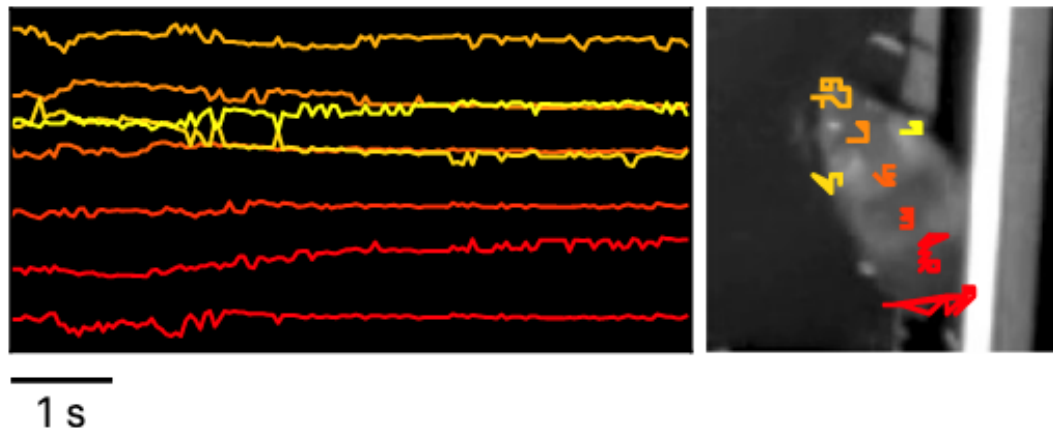
Users should select a kappa that best matches the mean of the changepoints distribution

# Deal with jittering of markers

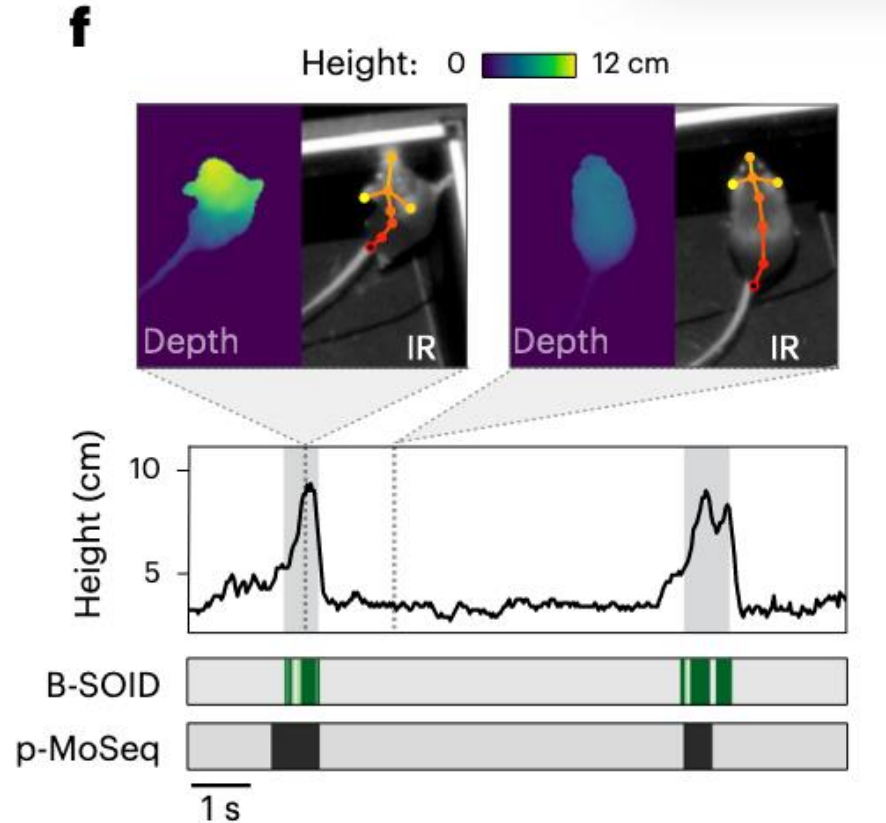
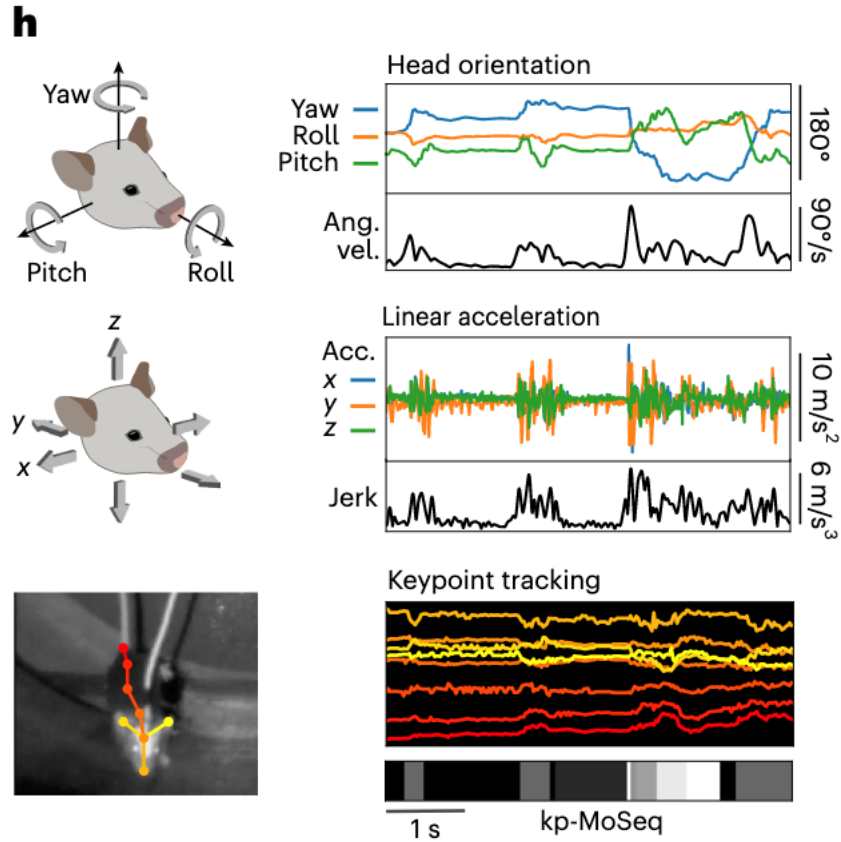
**a**



Keypoint jitter while mouse is motionless



# Keypoint-Moseq can work with 2D or 3D data



# Dataset preparation

## How many/which keypoints?

A fine starting point is 5-10 keypoints. For rodents, we recommend omitting the tail. The most important aspect is that the keypoints are informative and provide a holistic description of the animal's pose. If you are already tracking 4 keypoints along the spine, for example, adding 4 more may not add much new information. Note that it is always possible to exclude keypoints from modeling using the `use_bodyparts` setting in the config.



# No need to preprocess

## **Should I preprocess keypoints?**

For 2D keypoint tracking we generally don't recommend this. Keypoint-MoSeq is designed to work with raw keypoints. Preprocessing (e.g., smoothing, filtering, etc.) may remove important information.

# Input

Keypoint-MoSeq can be used with any method that produces 2D or 3D keypoint detections. Currently we support SLEAP, DeepLabCut, anipose, SLEAP-anipose, Neurodata Without Borders (NWB), Facemap, FreiPose and DANNCE. For methods not on this list, you can write a custom loading function or get in touch and request it as a new feature.

- If using one of the supported formats, data can be loaded as follows, optionally replacing `'deeplabcut'` with one of the following: `'sleap'`, `'anipose'`, `'sleap-anipose'`, `'nwb'`, `'facemap'`, `'freipose'`, `'DANNCE'`. The file formats expected in each case are described in the docstring for `keypoint_moseq.io.load_keypoints()`.

# Output: Syllables

- target syllable duration: rodents: ~400ms

